Koneru Lakshmaiah Education Foundation
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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
Faculty Name	Dr. Sukanya
Student Name	S. Ketana Sindhu
Roll Number	2520030487
Branch	CSE
Course Outcome	4

Case Study Topic: Pune-Mumbai Expressway Dynamic Toll Routing using Dijkstra Algorithm

Word Done Summary:

In this case study, we analysed dynamic toll routing on the Pune-Mumbai Expressway using Dijkstra's shortest path algorithm. Toll plazas were represented as graph vertices and toll charges were represented as weighted directed edges. We computed the minimum toll-cost path between KIR and PUN and studied how dynamic edge-weight updates affect shortest path computation. We also examined incremental graph updates and re-routing strategies for dynamic transportation systems.

Implementation Details:

1. Modelled toll plazas as graph vertices.
2. Represented toll costs using weighted directed edges.
3. Applied Dijkstra's shortest path algorithm.
4. Maintained predecessor tracking for path reconstruction.
5. Analysed dynamic edge-weight updates.
6. Compared full recomputation and incremental update strategies.
7. Evaluated routing efficiency in dynamic road networks.

Result:

Dijkstra's algorithm successfully computed the minimum toll-cost path on the Pune-Mumbai Expressway. Dynamic edge updates demonstrated how changing traffic conditions affect routing decisions in transportation networks.

Output:

```
Minimum toll costs from KIR:
Node 0 : 0
Node 1 : 12
Node 2 : 15
Node 3 : 30
Node 4 : 29
Node 5 : 45
Node 6 : 55
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GitHub Link:

https://github.com/ketanasindhusomisetty/DSA-Case-Studies/tree/main

Student Signature	Faculty Signature
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